
 $^{35}\text{Cl}(\text{p,d})$ **1972Vi04**

$^{35}\text{Cl} \rightarrow 1\text{n} + ^{34}\text{Cl}$ from $J^\pi = 3/2^+$ ^{35}Cl ground state.

1972Vi04: a 40-MeV proton beam was produced from the Grenoble variable-energy cyclotron. The target was chlorine gas of natural isotopic abundance. Deuterons were detected using a silicon telescope consisted of a 500- μm surface barrier ΔE detector, a 3-mm lithium-drifted silicon E detector, and another 3-mm lithium-drifted silicon E detector in anti-coincidence (FWHM=85 keV). Measured $\sigma(E_d, \theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

 ^{34}Cl Levels

Spectroscopic factor of is obtained by: $\sigma(\theta)_{\text{exp}} = 1.53[(2s_d+1)/(2s_p+1)][(2s_n+1)/(2(2L+1))]$ $C^2S\sigma(\theta)_{\text{DWBA}}$, where s_d , s_p , and s_n are the spins of the deuteron, proton, and neutron, respectively. 1.53 is a normalization factor related to the use of Hulthen wavefunction for the deuteron (**1966Ba54**). The overall uncertainty in C^2S is estimated to be about 30% (**1972Vi04**).

$E(\text{level})^\dagger$	$L^\dagger$	$C^2S^\dagger$	Comments
0.0	2	0.45	
140 15	2	1.20	
450 15	2	0.28	
670 15	0	0.16	
1230 15	0	0.50	
1880 15	0	0.16	
2160 15	0	0.32	
2600 20			E(level): doublet (1972Vi04).
3360 20	0	0.75	
4160 20	0	0.30	E(level): possibly multiplet (1972Vi04).
5000 30	2	1.70	E(level): possibly multiplet (1972Vi04).

† From **1972Vi04**.